

Laboratorio Modulo 4

Soluzioni

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Calcolo di eventi di probabilit  su distribuzioni di probabilit  parametriche

In questo laboratorio sar  richiesto di applicare le apposite funzioni di R per eseguire calcoli di probabilit  di eventi associati a variabili casuali continue distribuite secondo distribuzioni di densit  parametriche.

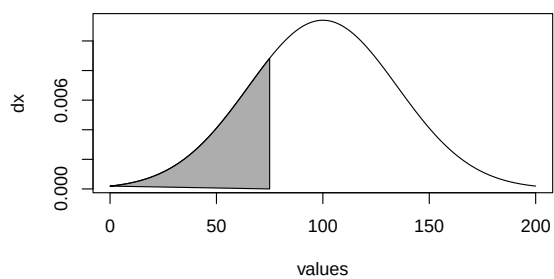
Esercizio 1

Sia X una variabile casuale continua distribuita secondo una funzione di densit  normale $N(x | \mu_X, \sigma_X^2)$ dove $\mu_X = 100$ e $\sigma_X = 35$. Si considerino i seguenti quattro eventi associati ad X : $E_1 = \{x \in \mathbb{R} : x \leq 75\}$, $E_2 = \{x \in \mathbb{R} : 50 \leq x \leq 70\}$, $E_3 = \{x \in \mathbb{R} : x \geq 110\}$, $E_4 = \{x \in \mathbb{R} : x \geq 125\}$. Si calcolino le probabilit  dei seguenti eventi combinati:

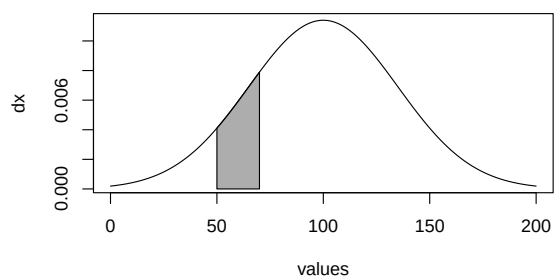
1. $P[E_1 \cup E_3]$
2. $P[E_1 \cup E_4]$
3. $P[E_1 \cap E_2]$
4. $P[E_1 \cap E_3]$.

```
pe1=pnorm(75,mx,sx)
pe2 = pnorm(70, mx, sx)- pnorm(50, mx, sx)
pe3 = 1- pnorm(110, mx, sx)
pe4 = 1- pnorm(125, mx, sx)

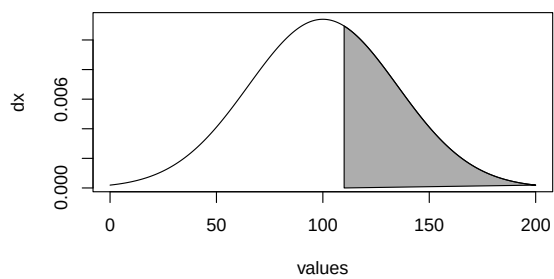
# unione e1 ed e3
plot(values, dx, type = "l")
polygon(c(values[values <= 75], 75), c(dx[values<= 75], 0), col = rgb(.2, .2,.2, alpha = .4))
polygon(c(values[values >= 110], 110), c(dx[values>= 110], 0), col = rgb(.2, .2,.2, alpha = .4))
```



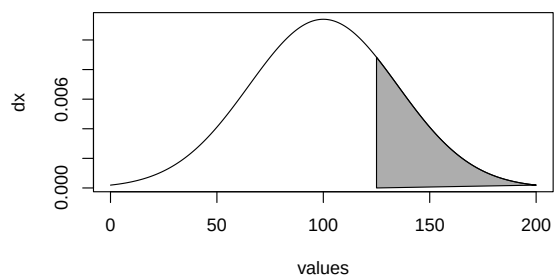
(a) E1



(a) E2

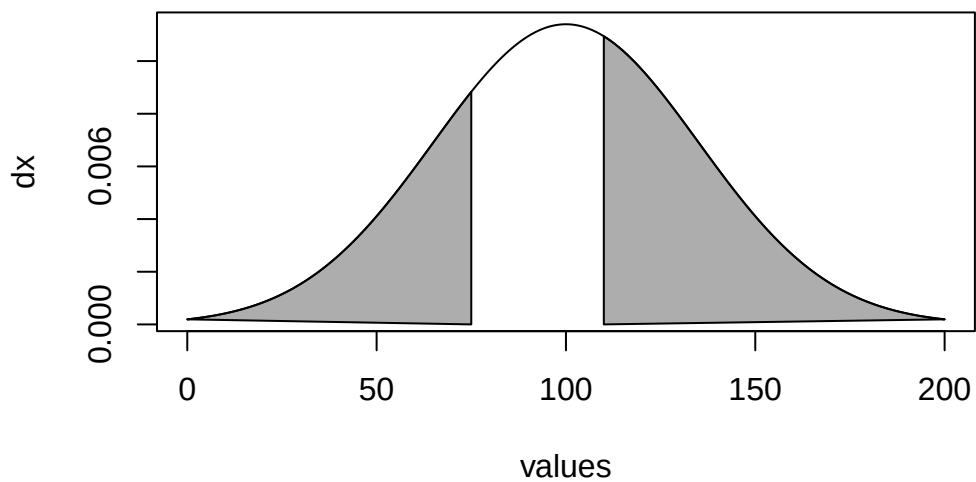


(a) E3



(a) E4

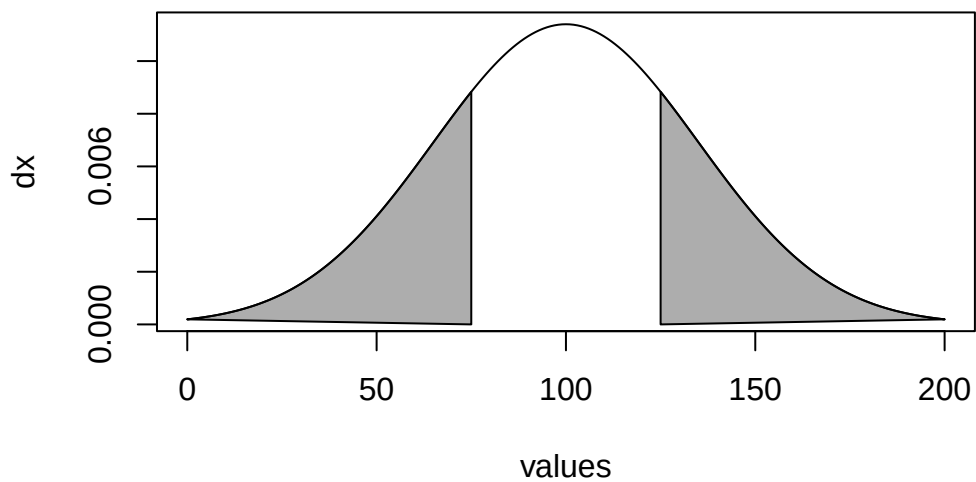
Illustrazione degli eventi relativi all'esercizio 1



```
pe1 + pe3
```

```
[1] 0.6250737
```

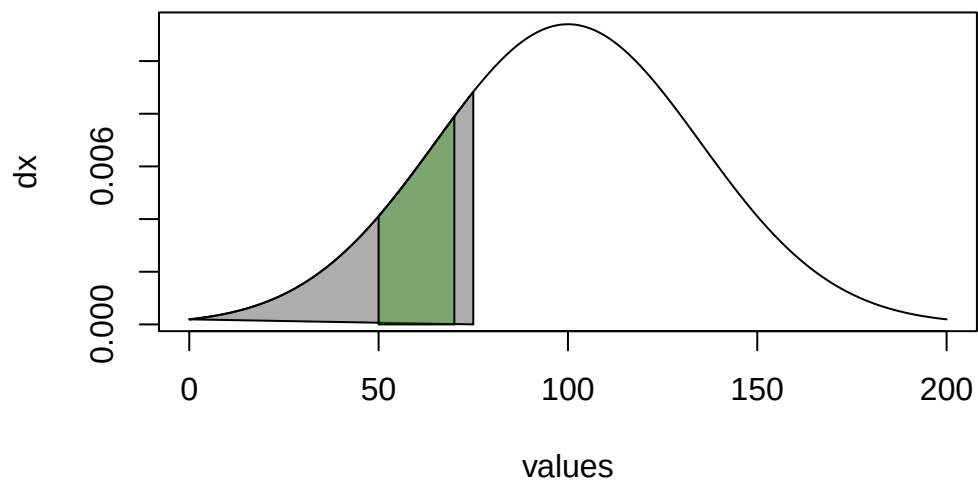
```
# 1.2: unione e1 e4
plot(values, dx, type = "l")
polygon(c(values[values <= 75], 75), c(dx[values<= 75], 0), col = rgb(.2, .2,.2, alpha = .4))
polygon(c(values[values >= 125], 125), c(dx[values>= 125], 0), col = rgb(.2, .2,.2, alpha =
```



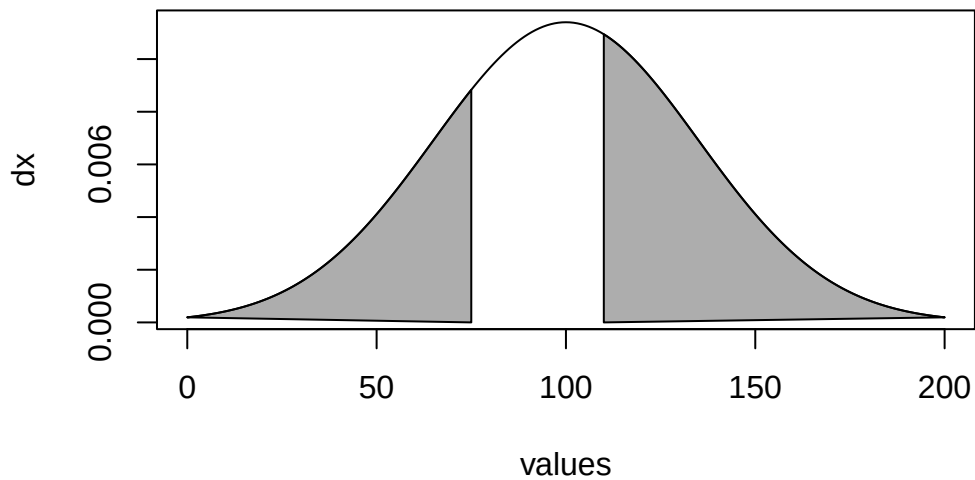
```
pe1 + pe4
```

```
[1] 0.4750505
```

```
# 1.3: intersezione e1, e2
plot(values, dx, type = "l")
polygon(c(values[values <= 75], 75), c(dx[values<= 75], 0), col = rgb(.2, .2,.2, alpha = .4))
polygon(c(50, values[values >= 50 & values <= 70], 70), c(0, dx[values >= 50 & values <= 70]
```



```
# 1.4 intersezione e1 e3
plot(values, dx, type = "l")
polygon(c(values[values <= 75], 75), c(dx[values<= 75], 0), col = rgb(.2, .2,.2, alpha = .4))
polygon(c(values[values >= 110], 110), c(dx[values>= 110], 0), col = rgb(.2, .2,.2, alpha = .4))
```

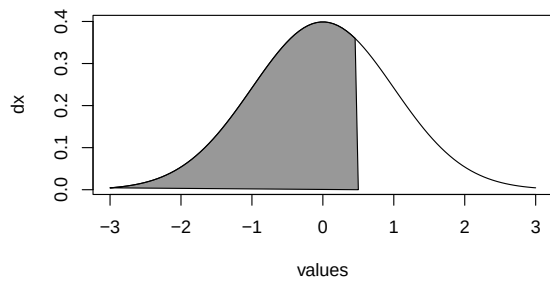


Esercizio 2

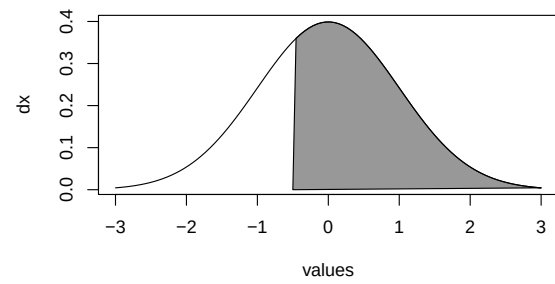
Sia X una variabile casuale continua distribuita secondo una funzione di densità normale standardizzata $N(x|\mu_X, \sigma_X)$ (dove chiaramente $\mu_X = 0$ e $\sigma_X = 1$). Si considerino i seguenti due eventi associati ad X : $E_1 = \{x \in \mathbb{R} : x \leq 0.5\}$ e $E_2 = \{x \in \mathbb{R} : x \geq -0.5\}$. Si calcolino le probabilità dei seguenti eventi combinati:

1. $P[E_1 \cup E_2]$.
2. $P[E_1 \cap E_2]$.
3. $P[E_1 \triangle E_2]$.

```
# esercizio 2.1: unione e1 ed e2
plot(values, dx, type = "l")
polygon(c(values[values <= 0.5], 0.5), c(dx[values <= 0.5], 0), col = rgb(.2, .2, .2, alpha = 0.5))
polygon(c(values[values >= -0.5], -0.5), c(dx[values >= -0.5], 0), col = rgb(.1, .1, .1, alpha = 0.5))
```

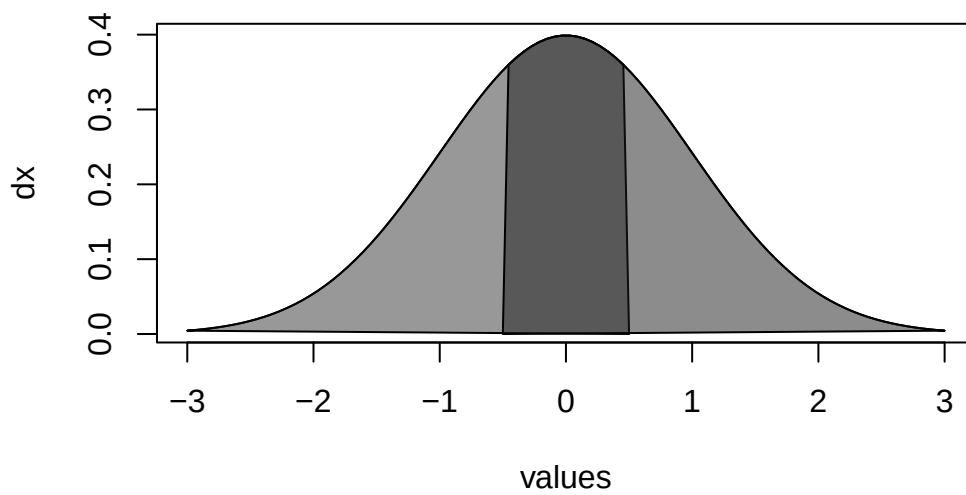


(a) E1



(a) E2

Illustrazione degli eventi relativi all'esercizio 2



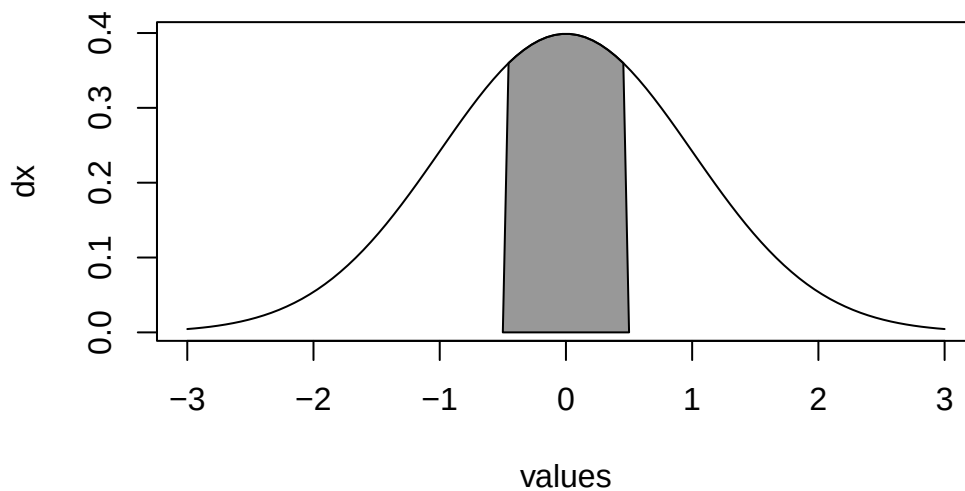
```
pe1 = pnorm(0.5)
pe2 = 1- pnorm(-0.5)
pe1 + pe2
```

```
[1] 1.382925
```

```
pe1intere2 = pnorm(0.5)- pnorm(-0.5)
pe1 + pe2- pe1intere2
```

```
[1] 1
```

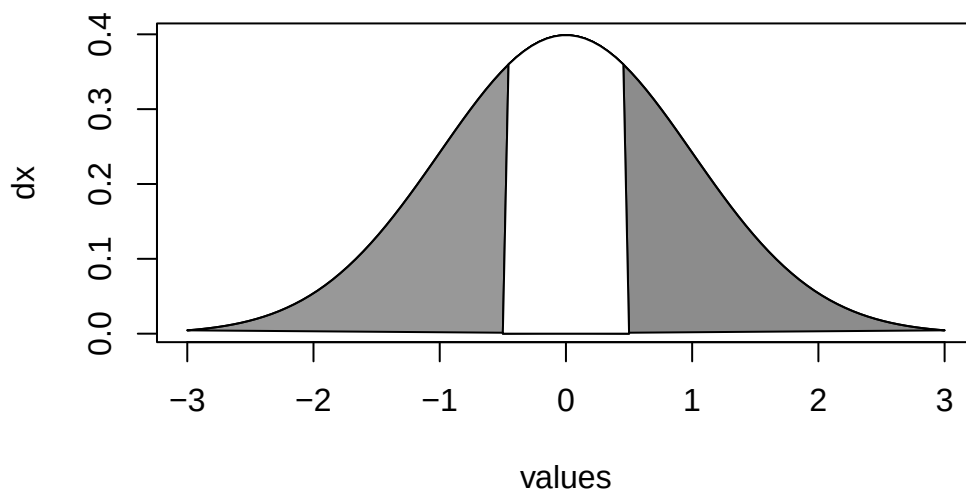
```
plot(values, dx, type = "l")
polygon(c(-0.5, values[values >=-0.5 & values <= 0.5], 0.5), c(0, dx[values >=-0.5 & values <= 0.5]), col = "gray", lty = 1)
```



```
pnorm(0.5)- pnorm(-0.5)
```

```
[1] 0.3829249
```

```
# esercizio 2.3 differenza simmetrica e1 ed e2
plot(values, dx, type = "l")
polygon(c(values[values <= 0.5], 0.5), c(dx[values <= 0.5], 0), col = rgb(.2, .2, .2, alpha = 0.5), lty = 1)
polygon(c(values[values >=-0.5], -0.5), c(dx[values >=-0.5], 0), col = rgb(.1, .1, .1, alpha = 0.5), lty = 1)
polygon(c(-0.5, values[values >=-0.5 & values <= 0.5], 0.5), c(0, dx[values >=-0.5 & values <= 0.5]), col = "gray", lty = 1)
```



```
pnorm(-0.5) + (1-pnorm(0.5))
```

```
[1] 0.6170751
```

Esercizio 3

Siano X ed Y due variabili casuali continue distribuite secondo le seguenti funzioni di densità normale: $N(x|\mu_X, \sigma_X)$ (con $\mu_X = 10$ e $\sigma_X = 5$) e $N(y|\mu_Y, \sigma_Y)$ (con $\mu_Y = 7$ e $\sigma_Y = 8$). Inoltre si assuma che le due variabili casuali X ed Y siano anche indipendenti, cioè a dire:

$$P(X, Y) = P(X)P(Y)$$

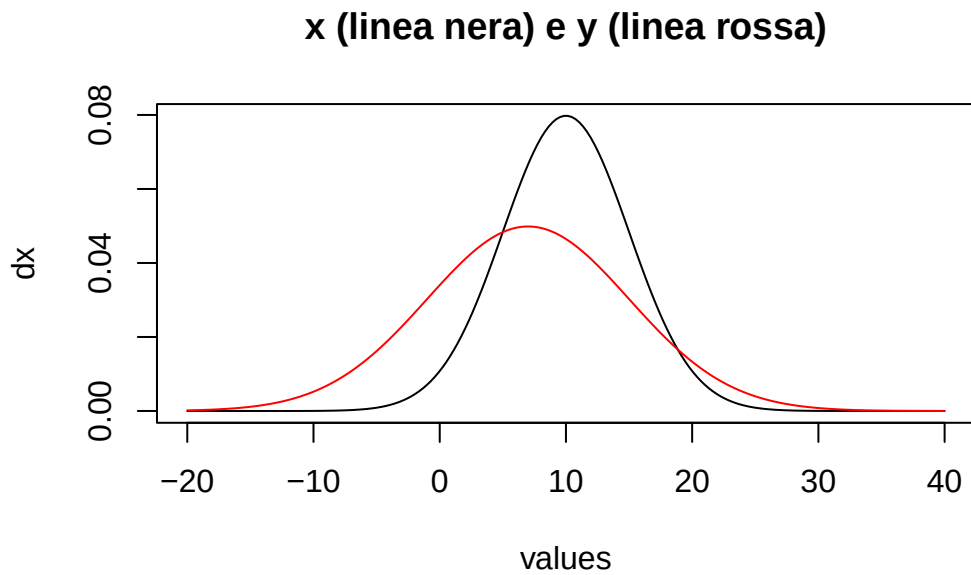
Si considerino i seguenti due eventi associati ad X : $E_1 = \{x \in \mathbb{R} : 5 \leq x \leq 9\}$ e $E_2 = \{x \in \mathbb{R} : 6.5 \leq x \leq 13\}$. Per Y si ha un solo evento: $G_1 = \{y \in \mathbb{R} : 4 \leq y \leq 8.5\}$. Si calcolino le probabilità dei seguenti eventi combinati:

1. $P[E_1 \cup G_1]$.
2. $P[E_2 \cap G_1]$.

```

mx = 10
sx = 5
my = 7
sy = 8
values = seq(-20, 40, length.out = 200)
dx = dnorm(values, mx, sx)
dy = dnorm(values, my, sy)
plot(values, dx, type = "l", main = "x (linea nera) e y (linea rossa)")
lines(values, dy, type = "l", col = "red")

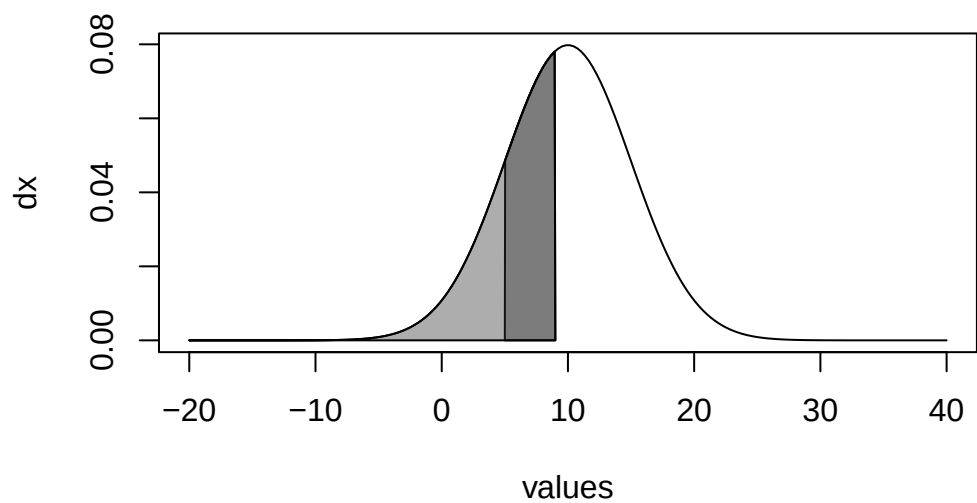
```



```

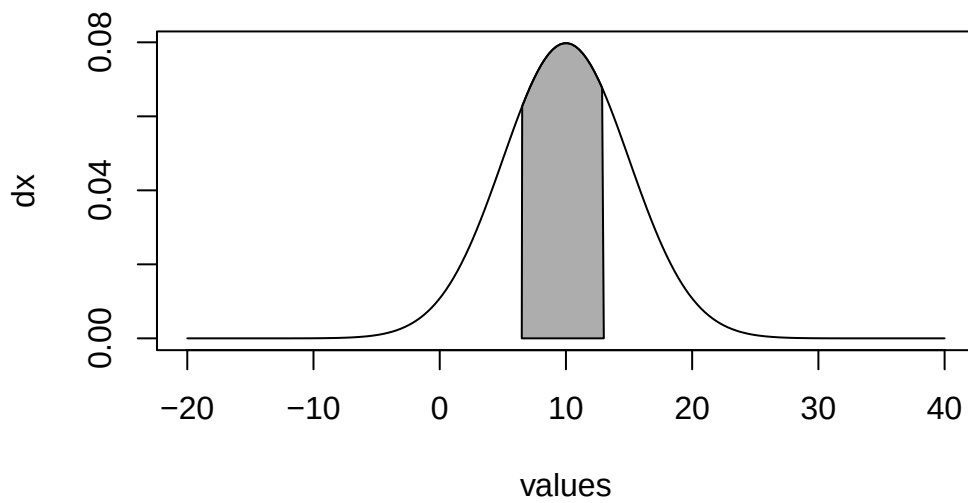
# e 1 sulla x
plot(values, dx, type = "l")
polygon(c(5, values[values>=5 & values <= 9], 9), c(0, dx[values>=5 & values <= 9], 0), col = "red")
polygon(c(values[values <= 9], 9), c(dx[values <= 9], 0), col = rgb(.2,.2,.2, alpha = .4))

```



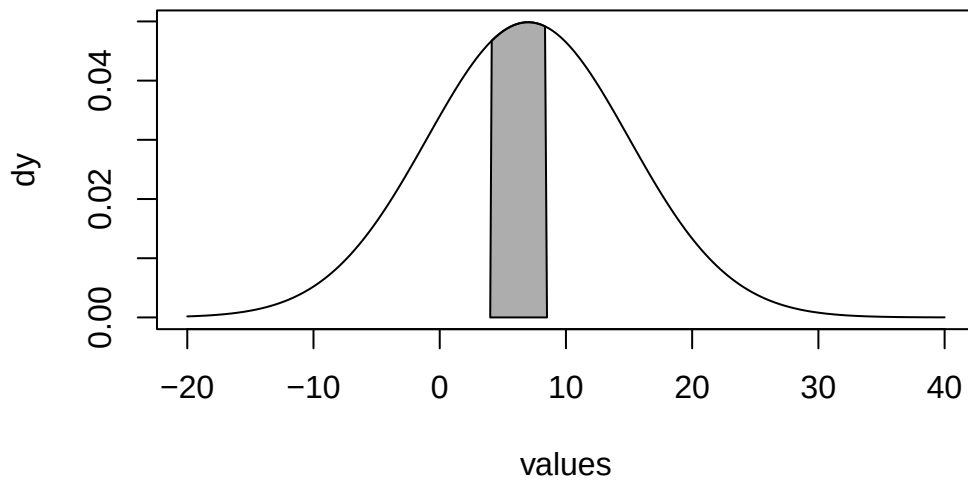
```
e1 = pnorm(9, mx, sx)- pnorm(5, mx, sx)

# e2 6.5 leq x leq 13
plot(values, dx, type = "l")
polygon(c(6.5, values[values>=6.5 & values <= 13], 13), c(0, dx[values>=6.5 & values <= 13],
```



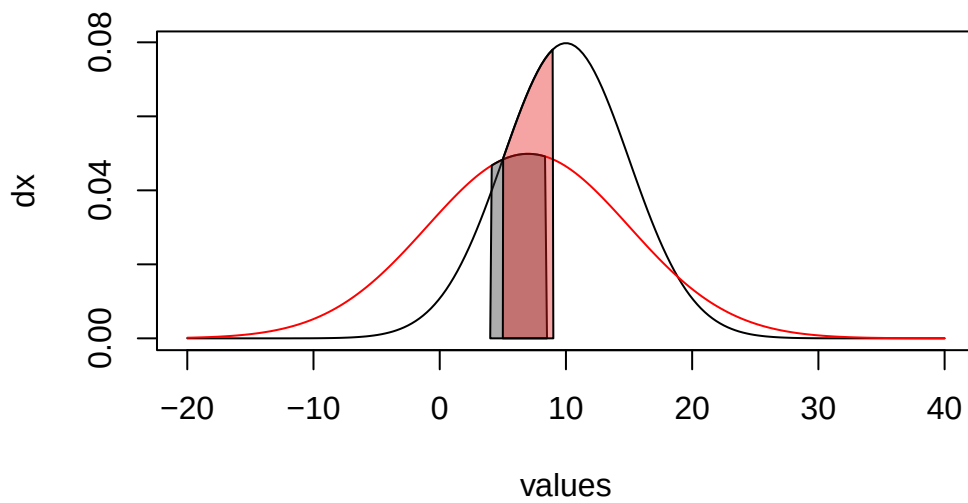
```
e2 = pnorm(13, mx, sx) - pnorm(6.5, mx, sx)

# evento y
# g 4 \leq x \leq 8.5
plot(values, dy, type = "l")
polygon(c(4, values[values >= 4 & values <= 8.5], 8.5), c(0, dy[values >= 4 & values <= 8.5], 0))
```



```
g1 = pnorm(8.5, my, sy)- pnorm(4, my, sy)

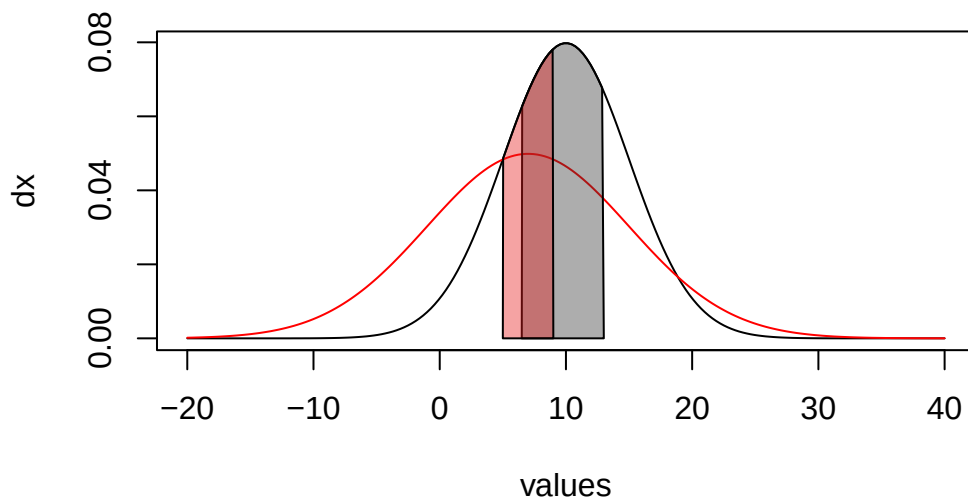
# 3.1:unione e1 e g1
plot(values, dx, type = "l")
lines(values, dy, type = "l", col = "red")
polygon(c(4, values[values>=4 & values <= 8.5], 8.5), c(0, dy[values>=4 & values <= 8.5], 0))
polygon(c(5, values[values>=5 & values <= 9], 9), c(0, dx[values>=5 & values <= 9], 0), col = "red")
```



```
e1 + g1- (e1*g1)
```

```
[1] 0.4248214
```

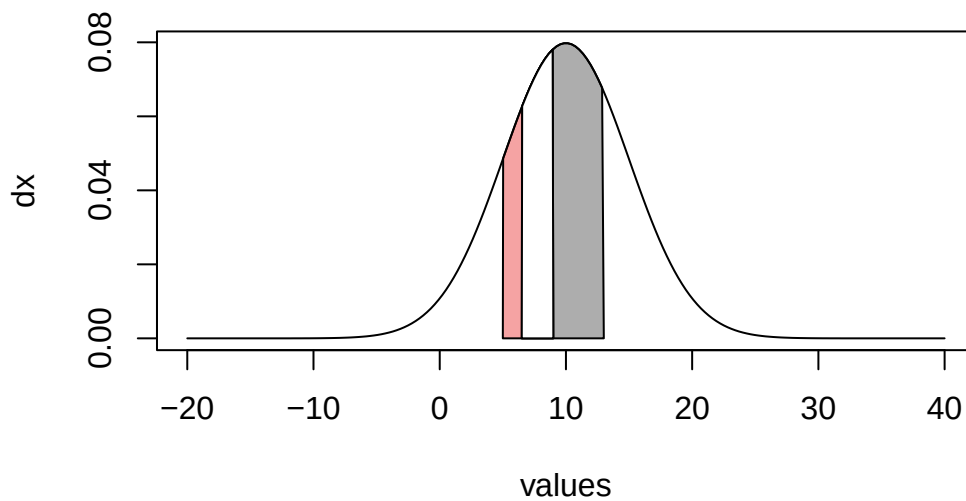
```
# 3.2 intesrezione e2 g1
plot(values, dx, type = "l")
lines(values, dy, type = "l", col = "red")
polygon(c(6.5, values[values>=6.5 & values <= 13], 13), c(0, dx[values>=6.5 & values <= 13],
polygon(c(5, values[values>=5 & values <= 9], 9), c(0, dx[values>=5 & values <= 9], 0), col =
```



```
e2*g1
```

```
[1] 0.1066914
```

```
# differenza simmetrica evento e1 ed e2 riferiti alla variabile x
plot(values, dx, type = "l")
polygon(c(6.5, values[values>=6.5 & values <= 13], 13), c(0, dx[values>=6.5 & values <= 13],
polygon(c(5, values[values>=5 & values <= 9], 9), c(0, dx[values >=5 & values <= 9], 0), col
polygon(c(6.5, values[values>=6.5 & values <= 9], 9), c(0, dx[values>=6.5 & values <= 9], 0)
```



```
rosa = pnorm(6.5, mx, sx)- pnorm(5, mx, sx)
grigio = pnorm(13, mx, sx)- pnorm(9, mx, sx)
rosa + grigio
```

```
[1] 0.388315
```

Esercizio 4

Sia W una variabile casuale continua distribuita secondo una funzione di densità t di Student $t(w|gdl)$ (dove $gdl = 35$). Si considerino i seguenti valori dei quantili di W :

1. $q_W(0.33)$.
2. $q_W(0.75) + q_W(0.25)$.
3. $q_W(0.75) - q_W(0.25)$.
4. $q_W(0.975)$.

```
# Esercizio 4.1
qt(0.33,35)
```

```
[1] -0.4436867
```

```
# Esercizio 4.2
```

```
qt(0.75,35) + qt(0.25,35)
```

```
[1] 0
```

```
# Esercizio 4.3
```

```
qt(0.75,35) - qt(0.25,35)
```

```
[1] 1.363128
```

```
# Esercizio 4.4
```

```
qt(0.975,35)
```

```
[1] 2.030108
```

Esercizio 5

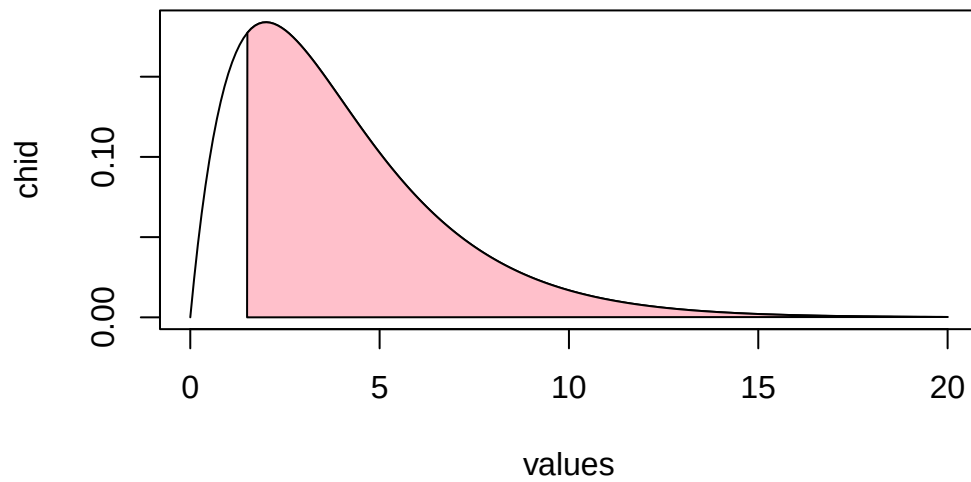
Sia U una variabile casuale continua distribuita secondo una funzione di densità Chi quadrato $\chi^2(u|gdl)$ ($gdl = 4$). Si calcolino le probabilità dei seguenti eventi di U :

1. $P[U \geq 1.5]$.
2. $P[1.1 \leq U \leq 2.5]$.
3. $P[U \geq q_U(0.950)]$.

```
values = seq(0, 20, length.out = 200)
chid = dchisq(values, 4)
plot(values, chid, type = "l")
```

```
# es 5. x >= 1.5
```

```
plot(values, chid, type = "l")
polygon(c(values[values >= 1.5], 1.5), c(chid[values >= 1.5], 0), col = "pink")
```



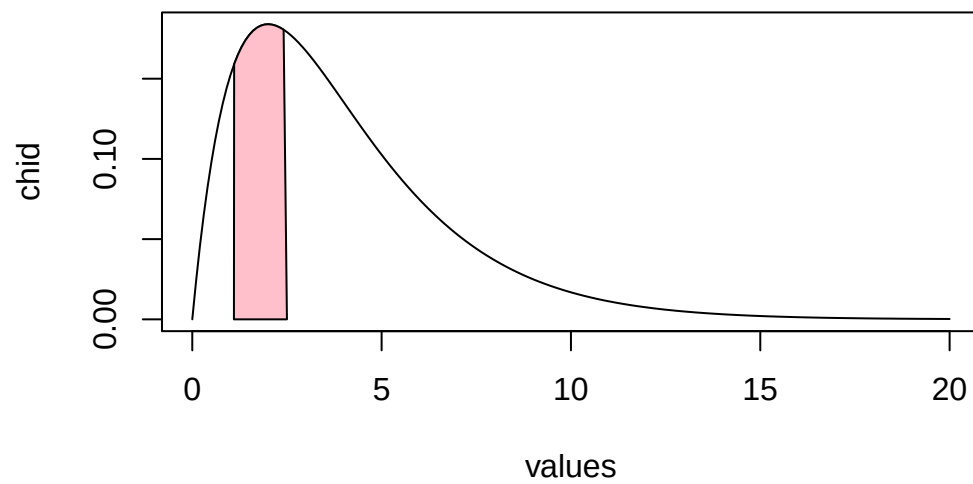
```
1- pchisq(1.5, 4)
```

```
[1] 0.8266415
```

```
# 5.2 1.1 \leq x \leq 2.5
```

```
plot(values, chid, type = "l")
```

```
polygon(c(1.1, values[values >= 1.1 & values <= 2.5], 2.5), c(0, chid[values >= 1.1 & values <= 2.5]), col = "pink", lty = 1)
```



```
pchisq(2.5, 4)- pchisq(1.1, 4)
```

```
[1] 0.2496364
```

```
# esercizio 5.3  
1- pchisq(qchisq(0.95, 4), 4)
```

```
[1] 0.05
```